

# Real Time Detection and Prediction of Water Contamination Level

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# Outline

- 1 Introduction
- 2 Motivation of the Work
- 3 State-of-The-Art Techniques
- 4 Research Gaps
- 5 Objectives
- 6 Proposed Methodology
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# Introduction

- Nature has blessed us with minimal fresh water resources.
- The continuous increase in population and their poor conduct have contributed to deteriorating water quality.
- It poses a threat to the health of its dependents.
- To monitor quality and rectify the causes behind it, we will employ IoT and ML techniques.



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# Motivation of the Work

- Limited natural resources need to be taken care of and made the most of it.
- In our region, traditional methods are still being employed, which are labor-intensive and time-consuming.
- This work involves IoT and ML, which paves the way to implement and enhance previous learning on the subject.



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# State-of-The-Art Techniques

- AloT using Simple recurrent unit(SRU).
- Fuzzy Similarity Analysis(FSA) enhancing ML estimates like Mixture Density Networks(MDN), Support Vector Regression(SVR) and Partial Least Squares Regression(PLSR).





# Literature Review

| S.No. | Author and Year                                                                                                   | Problem Statement                                                                                                                                                 | Material and Methods                                                                                                                | Results/Findings                                                                                                               | Limitations                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| [1].  | Wu-Chih Hu, Liang-Bi Chen, Bo-Hao Wang, Guo-Wei Li, and Xiang-Rui Huang, 2023                                     | Design and Implementation of a Full-Time Artificial Intelligence of Things-Based Water Quality Inspection and Prediction System for Intelligent Aquaculture       | Deployed an on site prototype for sample collection. Implemented the SRU model and evaluated the corresponding RMSE, MAPE, and MAE. | Deployed prototype was capable of handling algal decomposition. Performance of the SRU model was found to be better than LSTM. | More features could have been added to analyze performance.           |
| [2].  | Dehkordi, Alireza Taheri, Mohammad Javad Valadan Zoej, Ali Mehran, Mohsen Jafari, and Amir Masoud Chegolian, 2024 | Fuzzy similarity analysis of effective training samples to improve machine learning estimations of water quality parameters using Sentinel-2 remote sensing data. | Implementation of FSA to enhance the performance of several ML models.                                                              | Increased accuracy of MDN, SVR, and PLSR for estimating two target WQPs                                                        | Delayed data detection causes hindrance in analysing actual features. |



# Literature Review (contd.)

| S.No. | Author and Year               | Problem Statement                                                                                                | Material and Methods                                                                                                              | Results/Findings                                                                                                     | Limitations                                                                                                        |
|-------|-------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| [3].  | Manish Kumar.et.al, 2023      | Quality Assessment and Monitoring of River Water Using IoT Infrastructure                                        | PCA was utilized to obtain most valuable features in factor analysis and appropriate weights were assigned to concerned features. | WQI of Ganga was better than of Sangam, i.e., infrastructure is capable of identifying comparatively polluted zones. | Observation points could have been increased to get a more clear idea about sources responsible for contamination. |
| [4].  | Vandana K. Chalka.et.al, 2024 | Rapid and Sensitive Electrochemical Detection of Escherichia coli in water using Cr-Au IDE-Porous Silicon Sensor | Fabrication of porous silicon(PSi), synthesis of bacterial dilutions.                                                             | The response and recovery time of the sensor are 80 and 90ms respectively.                                           | Onsite testing is yet to be done.                                                                                  |



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# Research Gaps

- The number of features needs to be increased to obtain a detailed overview of the model's performance.
- The dependency on remote sensing causes a delay in the estimation of actual features.
- Need to increase the observation points to identify the crucial sources of contaminations.



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# Objectives

- Implement a well-organized prototype for data collection with relevant features.
- Formation of a data pipeline for storage and model training purposes.
- Approximately optimizing the model to obtain improved performance in information acquisition.
- Sharing information in real-time through the implementation and deployment of dedicated applications for the entities concerned.



# Classification Table

| Designated-Best-Use <sup>1</sup>                                            | Class of water | Criteria                                                                                                                                                                   |
|-----------------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drinking Water Source without conventional treatment but after disinfection | A              | Total Coliforms Organism MPN/100ml shall be 50 or less<br>pH between 6.5 and 8.5<br>Dissolved Oxygen 6mg/l or more<br>Biochemical Oxygen Demand 5 days 20°C 2mg/l or less  |
| Outdoor bathing (Organised)                                                 | B              | Total Coliforms Organism MPN/100ml shall be 500 or less<br>pH between 6.5 and 8.5<br>Dissolved Oxygen 5mg/l or more<br>Biochemical Oxygen Demand 5 days 20°C 3mg/l or less |
| Drinking water source after conventional treatment and disinfection         | C              | Total Coliforms Organism MPN/100ml shall be 5000 or less<br>pH between 6 to 8.5<br>Dissolved Oxygen 4mg/l or more<br>Biochemical Oxygen Demand 5 days 20°C 3mg/l or less   |
| Propagation of Wildlife and Fisheries                                       | D              | pH between 6.5 to 8.5<br>Dissolved Oxygen 4mg/l or more<br>Free Ammonia (as N) 1.2 mg/l or less                                                                            |
| Irrigation, Industrial Cooling, Controlled Waste disposal                   | E              | pH between 6.0 to 8.5<br>Electrical Conductivity at 25°C micro mhos/cm Max.2250<br>Sodium absorption Ratio Max. 26<br>Boron Max. 2mg/l                                     |

<sup>1</sup> [https://uppcb.up.gov.in/pdf/Designated\\_Best\\_Use\\_Water\\_Quality\\_Criteria.pdf](https://uppcb.up.gov.in/pdf/Designated_Best_Use_Water_Quality_Criteria.pdf)



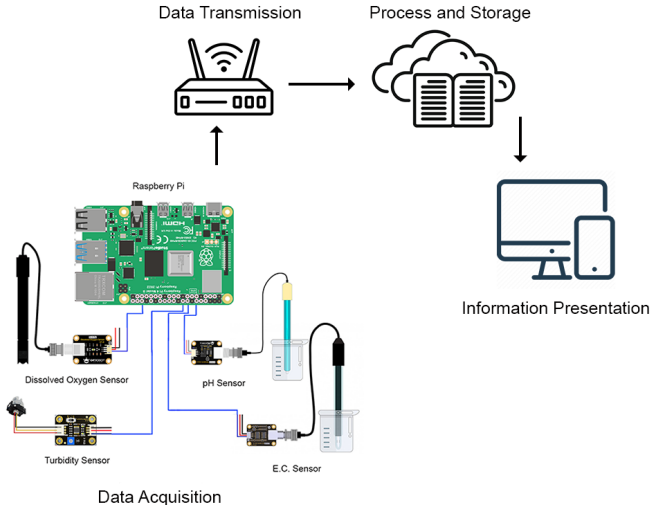
विद्या ददाति विनयं विनयाद् याति पात्रताम्

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# Proposed Architecture

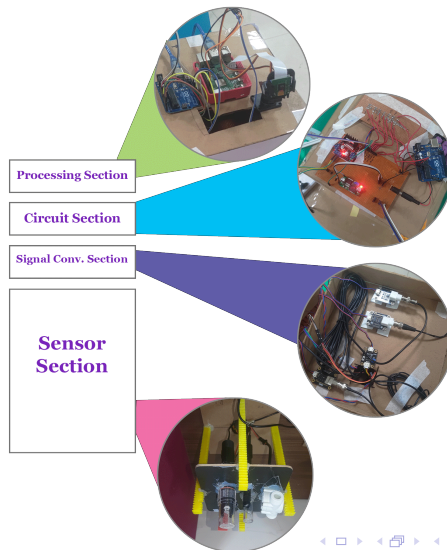


# Dataset Collection

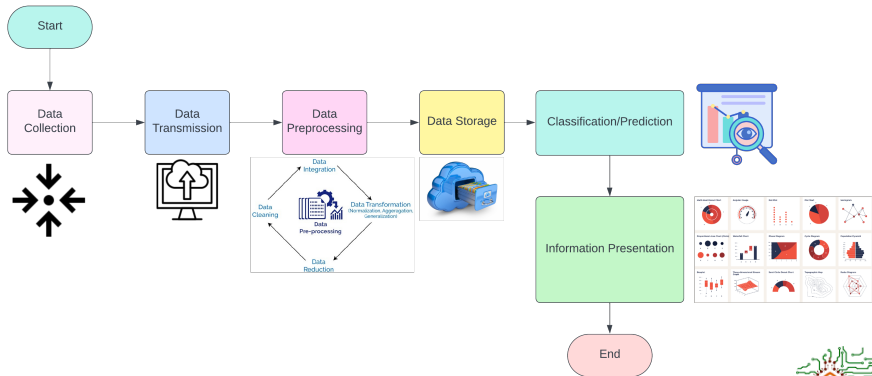
- As per the classification table, we need to collect data such as pH, dissolved oxygen, electrical conductivity, temperature etc.,
- We are implementing a buoyant prototype to generate additional features that can help in depicting the true nature of water bodies.
- We are also using sensors to collect flow rate, turbidity, and a light sensor to capture images.



# Hardware Organisation



# Classification Flowchart



# Estimation Technique

- To perform multiclass classification, we employ Random Forest Classification.
- We have used a camera to estimate bank's height from water level.
- For height estimation, we are using one reference object of known dimensions.
- We have trained a YOLOv8 model for object detection and we aim to find the minimal distance between two bounding boxes[5].



# Distance Estimation



Reference Object and Waterline

$$\text{Bank Depth (ft)} = \frac{\text{Waterline}_Y - \text{Reference Object Base}_Y}{\text{Pixels per Foot}}$$

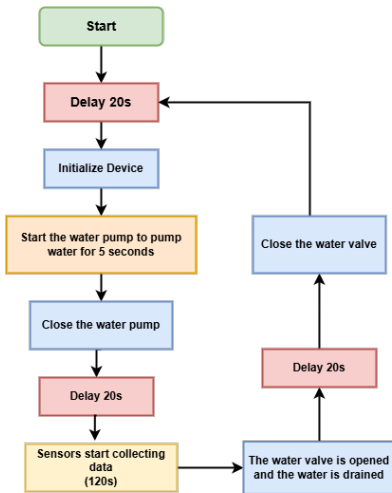
## Pixel-to-Foot Conversion Formula



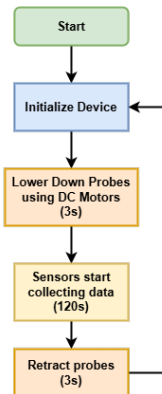
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# Operational Improvements



Flowchart of Wu-Chih Hu et al.

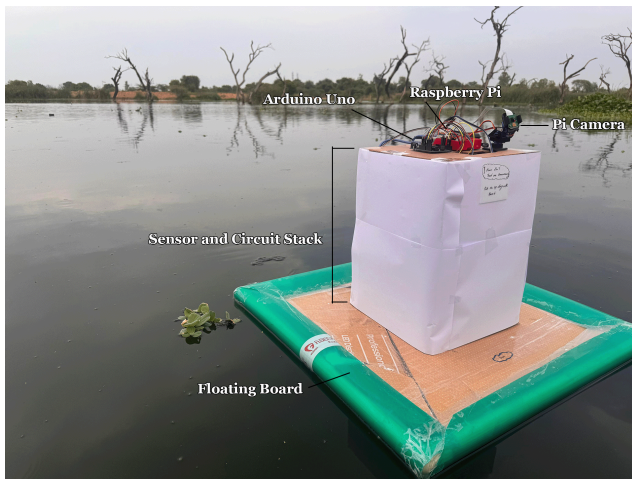


Flowchart of Proposed Prototype

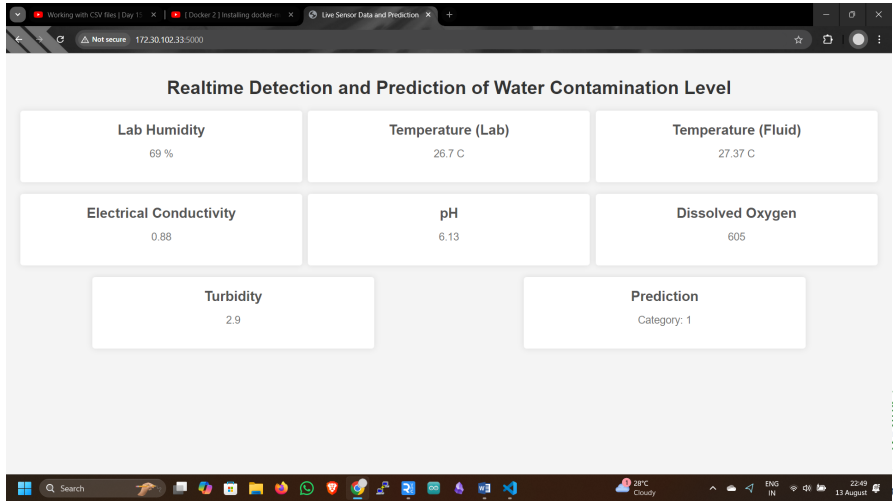




# In-situ Deployment

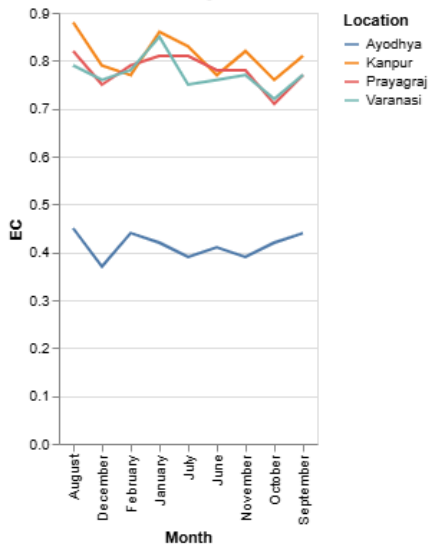


# Information Presentation

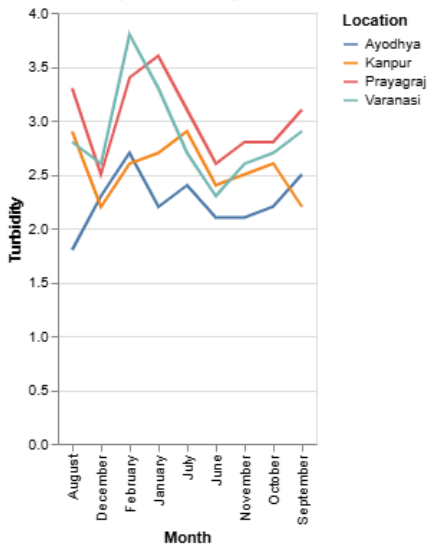


# Collected Dataset

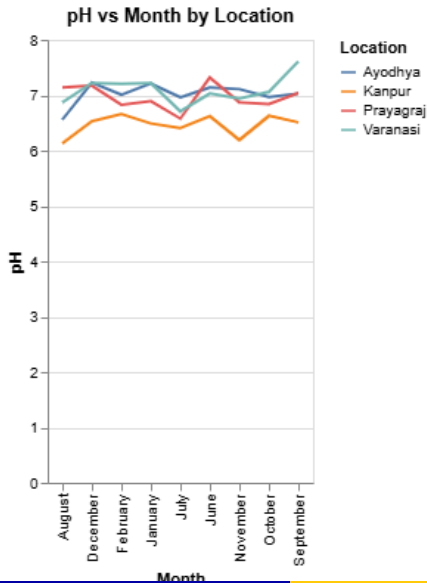
## EC vs Month by Location



## Turbidity vs Month by Location



# Collected Dataset (contd..)



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# Conclusion

- Sample collection for data generation in lab environment is in progress.
- Lab testing and on-site testing of designed prototype is in progress.
- Implementation of ML pipeline for efficient data management and modelling.



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# Future Works

- Increase dataset to improve model accuracy.
- Adoption of Transfer Learning technique in edge network for real-time decision making.





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# References I

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- [2] Alireza Taheri Dehkordi et al. “Fuzzy similarity analysis of effective training samples to improve machine learning estimations of water quality parameters using Sentinel-2 remote sensing data”. In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (2024).
- [3] Manish Kumar et al. “Quality assessment and monitoring of river water using IoT infrastructure”. In: *IEEE Internet of Things Journal* 10.12 (2023), pp. 10280–10290.



# References II

- [4] Vandana Kumari Chalka, Kamaljit Rangra, and Saakshi Dhanekar. “Rapid and Sensitive Electrochemical Detection of Escherichia coli in water using Cr-Au IDE-Porous Silicon Sensor”. In: *IEEE Sensors Letters* (2024).
- [5] Aji Bijaksana Abadi and Syahfrizal Tahcfulloh. “Digital image processing for height measurement application based on python OpenCV and regression analysis”. In: *JOIV: International Journal on Informatics Visualization* 6.4 (2022), pp. 763–770.



Thank You!